Zhang Bin

Ph.D. Candidate in Mechanical Engineering
Robotics and Machine Intelligence Laboratory (ROMI Lab)
Department of Mechanical Engineering
The Hong Kong Polytechnic University
Kowloon, Hong Kong SAR, China

me-bin.zhang@connect.polyu.hk Google Scholar ResearchGate
0000-0001-6860-1951 Personal Homepage
Last updated in August 2026

Education

- **Ph.D. Candidate in Mechanical Engineering**
  The Hong Kong Polytechnic University, Hong Kong SAR, China, Sep. 2021 – present
  (Supervisor: Dr. David Navarro-Alarcon)
- **Visiting Ph.D. Student**, Division of Decision and Control Systems
  KTH Royal Institute of Technology, Stockholm, Sweden, Mar. 2024 – Jul. 2024
  (Host advisor: Prof. Dimos V. Dimarogonas)
- **M.Sc. in Control Science and Engineering**
  China Academy of Space Technology, Beijing, China, Sep. 2017 – Jul. 2020
  (Supervisor: Academician WU Hongxin)
- **B.Eng. in Detection, Guidance, and Control Technology**
  Beihang University, Beijing, China, Sep. 2013 – Jul. 2017
  (FYP Advisor: Prof. LI Huifeng)

Employment

- **Research Assistant**, ROMI Lab
  The Hong Kong Polytechnic University, Hong Kong SAR, China, May 2021 – Aug. 2021
- **Research Intern**, Qian Xuesen Laboratory
  China Academy of Space Technology, China, Aug. 2020 – Mar. 2021

Research Fields

- My research interests lie in multi-agent systems, formation control, multi-robot manipulation, and control theory and applications.

Honors & Awards

- Outstanding Graduate of Beihang University, 2017
- National Inspirational Scholarship, 2015
- National Inspirational Scholarship, 2014

Teaching

- Teaching Assistant, Perceptual Robotics, 2021
- Teaching Assistant, Linear Systems and Control, 2022
- Teaching Assistant, Advanced Control Technology, 2022
- Teaching Assistant, Linear Systems and Control, 2023

Review Service

- ◇ Conference reviewer: IEEE International Conference on Robotics and Automation (ICRA), 2022 and 2024–2026; IEEE International Conference on Automation Science and Engineering (CASE), 2024.
- ◇ Journal reviewer: IEEE Control Systems Letters; Journal of Field Robotics; IEEE Robotics and Automation Magazine; IEEE Transactions on Control of Network Systems; Aerospace Science and Technology; IEEE Transactions on Robotics; IEEE Robotics and Automation Letters.

Skills

- ◇ **Languages:** Chinese (mother tongue), English.
- ◇ **Programming:** Python, C/C++, MATLAB, HTML, LaTeX.
- ◇ **Tools:** ROS, Linux, Webots, Arduino, SolidWorks.

Publication List

◇ **Journal Articles**
[#] Equal contribution

[J1] **B. Zhang**, H. Zhi, J. G. Romero, C. Yang, and D. Navarro-Alarcon, "Hug and Haul: Nonprehensile Object Transport by Deformable Tube-Connected Multi-Robot Systems," Under review, 2026.

[J2] X. Shao[#], **B. Zhang**[#], H. Zhi, J. G. Romero, B. Fan, Q. Hu, and D. Navarro-Alarcon, "InPTC: Integrated Planning and Tube-Following Control for Prescribed-Time Collision-Free Navigation of Wheeled Mobile Robots," *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 5, pp. 3339–3350, Oct. 2025. doi: 10.1109/TMECH.2024.3486727

[J3] **B. Zhang**, X. Shao, H. Zhi, L. Qiu, J. G. Romero, and D. Navarro-Alarcon, "Leader-Follower Formation Control of Perturbed Nonholonomic Agents along Parametric Curves with Directed Communication," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8603–8610, Oct. 2024. doi: 10.1109/LRA.2024.3445657

[J4] **B. Zhang**, H. Zhi, J. G. Romero, L. Labazanova, A. Duan, X. Li, and D. Navarro-Alarcon, "Fourier-Based Multi-Agent Formation Control to Track Evolving Closed Boundaries," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 11, pp. 4549–4559, Nov. 2023. doi: 10.1109/TCSI.2023.3300772

[J5] **B. Zhang** and J. Zhou, "Characteristic Model-Based Station-Keeping Control for Halo Orbit," *Acta Aeronautica et Astronautica Sinica*, vol. 40, no. 11, Nov. 2019. (in Chinese) doi: 10.7527/S1000-6893.2019.23206

[J6] Y. Tian, **B. Zhang**, X. Shao, and D. Navarro-Alarcon, "Encircling General 2-D Boundaries by Mobile Robots with Collision Avoidance: A Vector Field Guided Approach," *IEEE/ASME Transactions on Mechatronics*, Early Access, 2026. doi: 10.1109/TMECH.2026.3663573

[J7] N. Han, G. Gong, **B. Zhang**, Y. Xu, B. Yang, Y. Liu, and D. Navarro-Alarcon, "Prescribed Performance Control of Deformable Object Manipulation in Spatial Latent Space," *IEEE/ASME Transactions on Mechatronics*, Early Access, 2026. doi: 10.1109/TMECH.2026.3657722

[J8] H. Zhi, **B. Zhang**, J. Qi, J. G. Romero, X. Shao, C. Yang, and D. Navarro-Alarcon, "Non-Prehensile Object Transport by Nonholonomic Robots Connected by Linear Deformable Elements," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8651–8658, Oct. 2024. doi: 10.1109/LRA.2024.3440096

[J9] H. Y. Lee, P. Zhou, **B. Zhang**, L. Qiu, B. Fan, A. Duan, J. Tang, T. L. Lam, and D. Navarro-Alarcon, "A Distributed Dynamic Framework to Allocate Collaborative Tasks Based on Capability Matching in Heterogeneous Multi-Robot Systems," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 16, no. 1, pp. 251–265, Feb. 2024. doi: 10.1109/TCDS.2023.3264034

[J10] W. Ma, **B. Zhang**, L. Han, S. Huo, H. Wang, and D. Navarro-Alarcon, "Action Planning for Packing Long Linear Elastic Objects into Compact Boxes with Bimanual Robotic Manipulation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 3, pp. 1718–1729, Jun. 2023. doi: 10.1109/TMECH.2022.3223990

[J11] M. Muddassir, G. Limbert, **B. Zhang**, A. Duan, J.-J. Tan, and D. Navarro-Alarcon, "Model Predictive Thermal Dose Control of a Robotic Laser System to Automate Skin Photorejuvenation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 2, pp. 737–747, Apr. 2023. doi: 10.1109/TMECH.2022.3218806

[J12] S. Huo, **B. Zhang**, M. Muddassir, D. T. W. Chik, and D. Navarro-Alarcon, "A Sensor-Based Robotic Line Scan System with Adaptive ROI for Inspection of Defects over Convex Free-Form Specular Surfaces," *IEEE Sensors Journal*, vol. 22, no. 3, pp. 2782–2792, Feb. 2022. doi: 10.1109/JSEN.2021.3132428

[J13] J. Zhou, J. Hu, Y. Bai, and **B. Zhang**, "Optimal Impulsive Time-Fixed Transfers around the Libration Points of the Restricted Three-Body Problem," *Astrophysics and Space Science*, vol. 365, pp. 1–18, May 2020. doi: 10.1007/s10509-020-03793-7

[J14] J. Zhou, J. Hu, and **B. Zhang**, "Approximate Analytical Solutions of Motion near the Collinear Libration-Points in Restricted Three-Body Problem," *Journal of Deep Space Exploration*, vol. 7, no. 1, Feb. 2020. (in Chinese) doi: 10.15982/j.issn.2095-7777.2020.20190408001

[J15] J. Zhou, J. Hu, and **B. Zhang**, "Analytical Solutions for Relative Motion in the Circular Restricted Three-Body Problem," *Journal of Astronautics*, vol. 41, no. 2, Feb. 2020. (in Chinese) doi: 10.3873/j.issn.1000-1328.2020.02.004

◇ **Conference Articles**

[C1] **B. Zhang**, C. Li, and J. Zhou, "Station-Keeping of Halo Orbits Using Convex Optimization-Based Receding Horizon Control," in *Proceedings of the 2021 China Automation Congress*, Oct. 2021, pp. 7160–7165. doi: 10.1109/CAC53003.2021.9728345

[C2] **B. Zhang** and J. Zhou, "Characteristic Model-Based Station-Keeping Control for Halo Orbits about Sun-Earth L1 Point," in *Proceedings of the 2019 Chinese Automation Congress*, Nov. 2019, pp. 418–423. doi: 10.1109/CAC48633.2019.8997351

[C3] A. Duan, **B. Zhang**, L. Hu, L. Feng, and D. Navarro-Alarcon, "Balancing Control of Wheeled-Legged Hexapedal Robots for Traversing Rough Terrains," in *Proceedings of the 2022 IEEE International Conference on Robotics and Biomimetics*, Dec. 2022, pp. 1905–1910. doi: 10.1109/ROBIO55434.2022.10011895